

# FIT AUDIT — Technology in Society

**Manuscript:** “Institutional Observability Under Scaled AI Governance”

**Date:** June 3, 2026 **Auditor:** Windsurf (autonomous)

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## Criterion 1: Scope Fit

**Assessment:** [GREEN] GREEN

The manuscript addresses AI governance at institutional scale — a core topic area for Technology in Society. The journal’s recent issues span AI ethics, algorithmic accountability, and governance. The paper’s focus on how deployment architecture creates structural oversight failure falls squarely within the journal’s “technology-society interactions” mandate.

**Evidence:** - Journal lists “AI ethics and AI governance” and “Algorithmic accountability and fairness” as recent topic areas - Paper examines institutional decision pipelines, compliance dashboards, and audit trails — all technology-society intersection points - The throughput-observability asymmetry is fundamentally a technology-society dynamic

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## Criterion 2: Topic Fit

**Assessment:** [GREEN] GREEN

The paper examines how AI deployment at scale degrades human oversight capacity — a technology-governance-society intersection. It covers: - Institutional governance under AI deployment - Human oversight pipeline saturation - Compliance framework inadequacy - Governance-by-design interventions

All are directly aligned with Technology in Society’s editorial scope.

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## Criterion 3: Novelty Fit

**Assessment:** [GREEN] GREEN

The paper introduces several novel contributions: 1. **Oversight saturation** as a distinct failure mode (differentiated from Meyer & Rowan’s institutional decoupling) 2. **SEDI index** — a computable governance observability metric from public record streams 3. **Kingman heavy-traffic analysis** applied to human-in-the-loop oversight pipelines ( $2.71\times$  faster saturation proof) 4. **Three governance-by-design principles** treating oversight capacity as a runtime constraint

These are original contributions, not incremental extensions. The journal values work that “enables decision-makers to thoughtfully engage” — SEDI directly enables this.

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## Criterion 4: Methodological Fit

**Assessment:** [YELLOW] YELLOW

**Strengths:** - Continuous state-space model with formal mathematical derivation - Monte Carlo simulation (120 runs, 95% CI) - Kingman’s G/G/1 heavy-traffic approximation - Calibration against published institutional data (Vanderbilt, Peterson Health)

**Concerns:** - The paper is primarily theoretical/modeling — Technology in Society typically expects empirical, qualitative, or mixed-methods work - No primary data collection (interviews, surveys, case studies) - Calibration anchors are from public analyses, not proprietary operational data (acknowledged in Limitations) - The paper reads more like a CS/AI conference paper than a social science journal article

**Mitigation:** - The journal accepts “theoretical” work if methodologically rigorous - The mathematical formalization is rigorous - The institutional cases (Vanderbilt, PHTI) provide real-world grounding - The paper explicitly positions itself as deriving governance-by-design interventions — applied implications

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## Criterion 5: Audience Fit

**Assessment:** [YELLOW] YELLOW

**Strengths:** - Technology in Society readers include policymakers, governance professionals, and interdisciplinary scholars - The paper’s governance-by-design implications directly address decision-makers - SEDI is designed for external auditors — a clear policy actor

**Concerns:** - The mathematical content (state-space models, Kingman approximations, log-normal service times) may be inaccessible to some Technology in Society readers - The paper assumes familiarity with queueing theory and computational governance concepts - The current writing style is closer to AAAI/NeurIPS workshop format than a social science journal

**Mitigation:** - The abstract and introduction are accessible - The three mechanisms are explained in prose, not just equations - The governance implications section is policy-facing - Revision can add more explanatory framing for non-technical readers

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## Criterion 6: Policy/Applied Implication Fit

**Assessment:** [GREEN] GREEN

This is the strongest fit criterion. The paper: - Names specific policy actors: external auditors, governance bodies, frontline reviewers, vendor contract negotiators - Provides actionable interventions: SEDI monitoring, backpressure mechanisms, capacity gating, enforcement-authority co-location - Addresses a specific governance failure pattern: artifact-count compliance frameworks structurally incapable of detecting saturation - The cover letter can explicitly name the policy actor (auditors, governance middleware engineers, institutional review boards)

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## Criterion 7: Evidence Base

**Assessment:** [YELLOW] YELLOW

**Strengths:** - Mathematical proofs (Kingman, state-space model) - Monte Carlo simulation with confidence intervals - Published institutional boundary conditions

**Concerns:** - No empirical validation of the model against real institutional data - No primary evidence from institutions experiencing saturation - The  $D(\rho)$  degradation function is parameterized from literature, not measured - The paper acknowledges these limitations honestly

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## Summary

| Criterion              | Rating          |
|------------------------|-----------------|
| Scope Fit              | [GREEN] GREEN   |
| Topic Fit              | [GREEN] GREEN   |
| Novelty Fit            | [GREEN] GREEN   |
| Methodological Fit     | [YELLOW] YELLOW |
| Audience Fit           | [YELLOW] YELLOW |
| Policy Implication Fit | [GREEN] GREEN   |
| Evidence Base          | [YELLOW] YELLOW |

**Overall:** [GREEN] SUBMITTABLE WITH REVISIONS

The manuscript is a strong fit for Technology in Society’s topical scope and policy-implication requirements. The primary risks are: 1. Methodological expectations (more empirical than theoretical) 2. Audience accessibility (mathematical content) 3. Evidence base (no primary data)

These are addressable through revision without changing the paper’s core contributions. The paper should be reframed with more social-science exposition, less CS-conference density, and explicit policy-actor naming throughout.

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## Recommendation

**SUBMIT** after revisions that: 1. Add explanatory framing for non-technical readers around mathematical sections 2. Expand the institutional case grounding (Vanderbilt, PHTI) with more narrative detail 3. Strengthen the policy-actor identification throughout (not just in implications section) 4. Add a “Practical Implications for Governance Practitioners” subsection 5. Consider adding a glossary for technical terms ( $\rho_{oc}$ , SEDI, Kingman approximation)